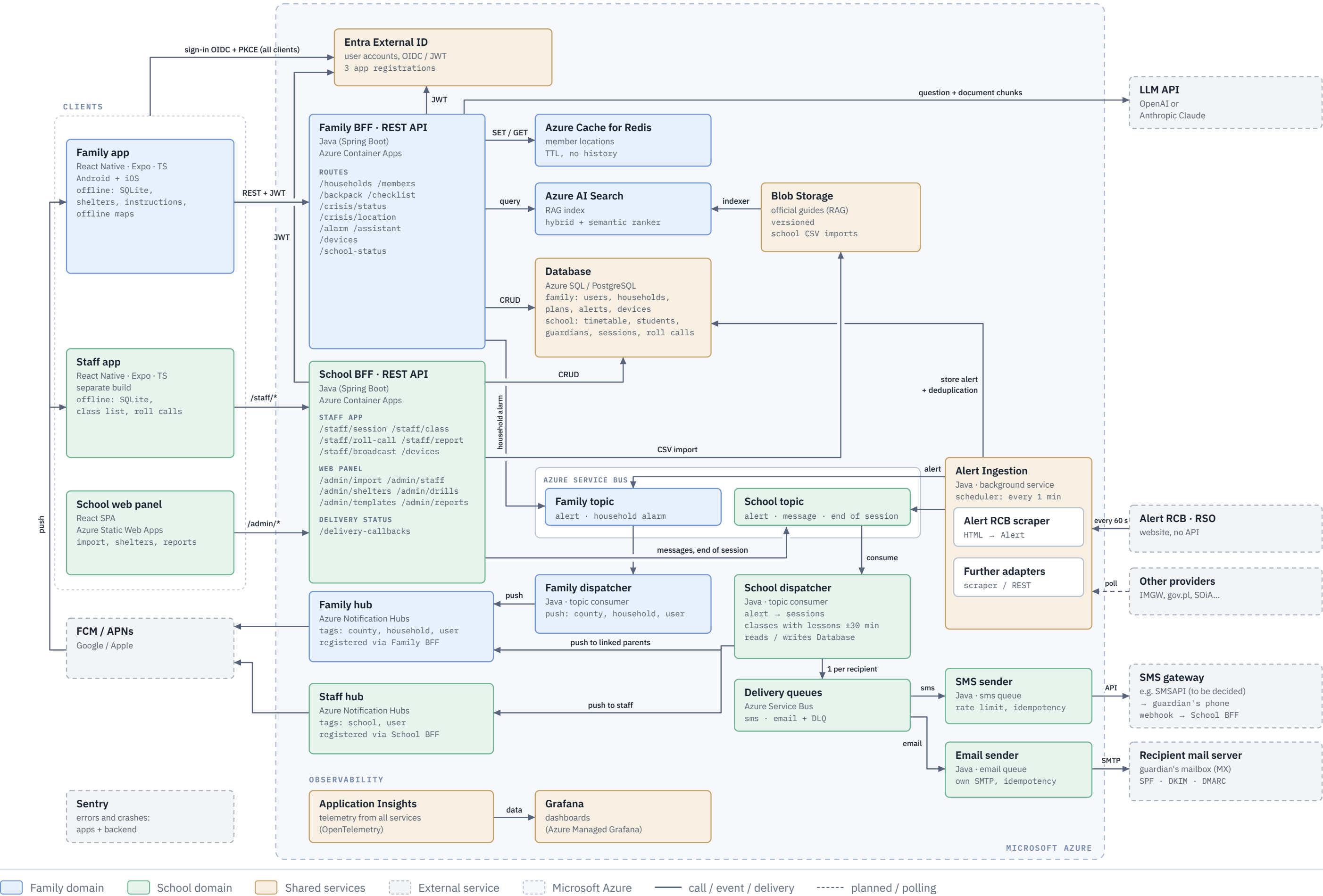


# Crisis Family Assistant

Family app, staff app and school web panel (React Native / React), two Java BFFs, everything on Azure. Box color shows the domain.



# Components

Every block on the diagram, grouped by domain.

| COMPONENT         | TECHNOLOGY                             | RESPONSIBILITY                                                                                                                                                                                                                                                                             |
|-------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLIENTS           |                                        |                                                                                                                                                                                                                                                                                            |
| Family app        | React Native (Expo), Android and iOS   | Household, go-bag checklist, family plan, crisis mode, child status from school. Shelters and instructions stored offline in SQLite.                                                                                                                                                       |
| Staff app         | React Native (Expo), separate build    | Teacher task, class roll call, director view with end of session, templated messages to parents. Class list and roll calls work offline.                                                                                                                                                   |
| School web panel  | React SPA, Azure Static Web Apps       | Import of students, guardians, timetable and staff. Shelters and routes, school calendar, drills, reports.                                                                                                                                                                                 |
| FAMILY DOMAIN     |                                        |                                                                                                                                                                                                                                                                                            |
| Family BFF        | Java, Spring Boot, Container Apps      | Only API for the family app: households, members, statuses, locations, household alarm, assistant. Reads child status from a read-only view of the school schema.                                                                                                                          |
| Redis             | Azure Cache for Redis                  | Last known member location with TTL, no history. Written only in crisis mode.                                                                                                                                                                                                              |
| AI Search         | Azure AI Search                        | Hybrid RAG index over official guides from Blob Storage. The LLM answers only from returned chunks and cites the source.                                                                                                                                                                   |
| Family topic      | Azure Service Bus topic                | Events: official alert, household alarm. Status changes are not published.                                                                                                                                                                                                                 |
| Family dispatcher | Java, topic consumer                   | Resolves recipients and sends push to the Family hub (tags: county, household, user).                                                                                                                                                                                                      |
| Family hub        | Azure Notification Hubs                | Device registrations of the family app; delivers push through FCM and APNs.                                                                                                                                                                                                                |
| SCHOOL DOMAIN     |                                        |                                                                                                                                                                                                                                                                                            |
| School BFF        | Java, Spring Boot, Container Apps      | API for the staff app and web panel (/staff, /admin, delivery callbacks). Stores roll calls; publishes messages and end of session.                                                                                                                                                        |
| School topic      | Azure Service Bus topic                | Events: official alert, message to parents, end of session. Roll calls are not published.                                                                                                                                                                                                  |
| School dispatcher | Java, topic consumer                   | On alert: finds classes with lessons now (Mon–Fri, 30 min buffer), opens sessions, pushes to staff, sends SMS and email to guardians, pushes to linked parent accounts.                                                                                                                    |
| Delivery queues   | Azure Service Bus queues               | sms and email, one message per recipient, each with a dead-letter queue.                                                                                                                                                                                                                   |
| SMS sender        | Java, queue consumer                   | Sends through the SMS gateway API with rate limiting, retries and an idempotency key.                                                                                                                                                                                                      |
| Email sender      | Java, queue consumer                   | Sends directly over SMTP to the recipient mail server (SPF, DKIM, DMARC), with retries and an idempotency key.                                                                                                                                                                             |
| Staff hub         | Azure Notification Hubs                | Device registrations of the staff app (tags: school, user).                                                                                                                                                                                                                                |
| SHARED            |                                        |                                                                                                                                                                                                                                                                                            |
| Database          | Azure SQL or PostgreSQL (one instance) | One database with two schemas. Family: users, households, invitations, checklists, plans, alerts, device tokens. School: schools, rooms, shelters, routes, classes, timetable, calendar, staff, students, guardians, links to family accounts, sessions, roll calls, messages, deliveries. |
| Entra External ID | Microsoft Entra External ID            | Sign-in for all clients (OIDC + PKCE). Staff and panel accounts by invitation only. Each BFF accepts tokens of its own clients.                                                                                                                                                            |
| Blob Storage      | Azure Blob Storage                     | Versioned official guides for the RAG index and CSV files imported from schools.                                                                                                                                                                                                           |
| Alert Ingestion   | Java, scheduler every 1 min            | Polls alert providers through adapters, normalizes, stores each alert in the database (deduplication) and publishes it to both topics.                                                                                                                                                     |
| Observability     | App Insights, Grafana, Sentry          | Traces and metrics over OpenTelemetry, dashboards (alert polling, dead-letter queues, delivery reach), errors in Sentry.                                                                                                                                                                   |
| EXTERNAL          |                                        |                                                                                                                                                                                                                                                                                            |
| LLM API           | OpenAI or Anthropic Claude             | Generates assistant answers from retrieved document chunks.                                                                                                                                                                                                                                |
| Alert providers   | Alert RCB (RSO), IMGW, gov.pl, SOiA    | Official alert sources; scraper or REST adapter per provider.                                                                                                                                                                                                                              |
| SMS gateway       | e.g. SMSAPI (to be decided)            | Delivers SMS to guardians and reports delivery status via webhook to the School BFF.                                                                                                                                                                                                       |
| FCM / APNs        | Google, Apple                          | Deliver push notifications to devices, also when the app is closed.                                                                                                                                                                                                                        |